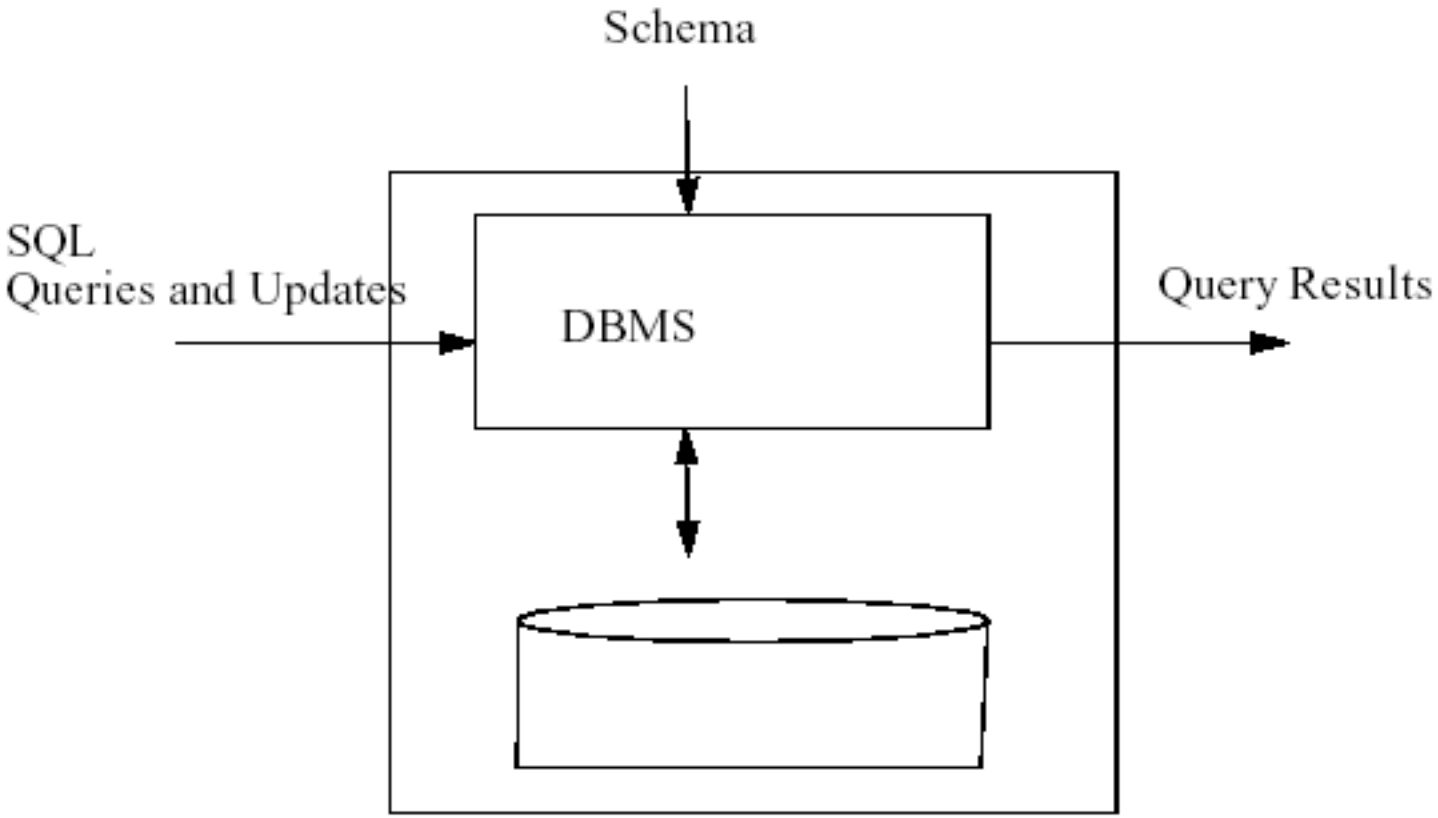
Active Databases

General principles of Conventional Database Systems



Conventional (Passive) Databases

Data model

Transaction model

- ACID principle

Examples of real world problems:

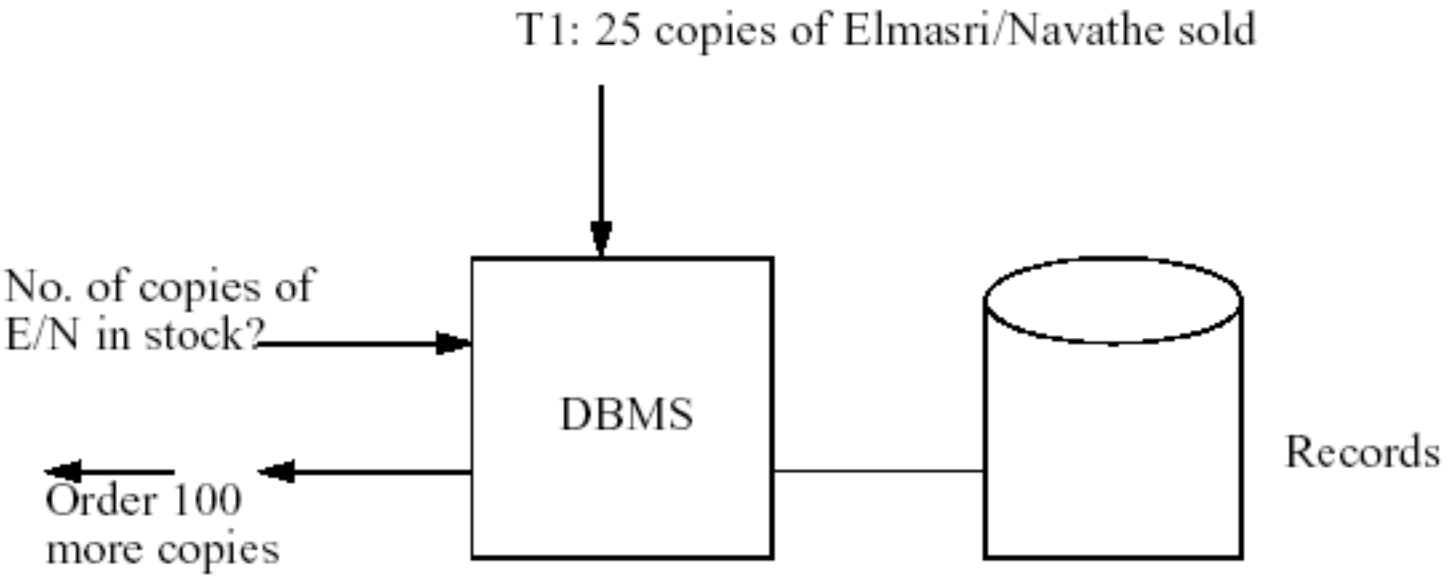
- Inventory control
  - reordering of items when quantity in stock falls below threshold.
- Travel waiting list
  - book ticket as soon as right kind is available
- Stock market
  - buy/sell stocks when price below/above threshold
- Maintenance of master tables (view materialization)
  - maintain table that contain sum of salaries for each department

Conventional Databases

Passive DBMS

Periodical polling of database by application

- Frequent polling => expensive
- Infrequent polling => might miss the right time to react
- DBMS does not know that application is polling.

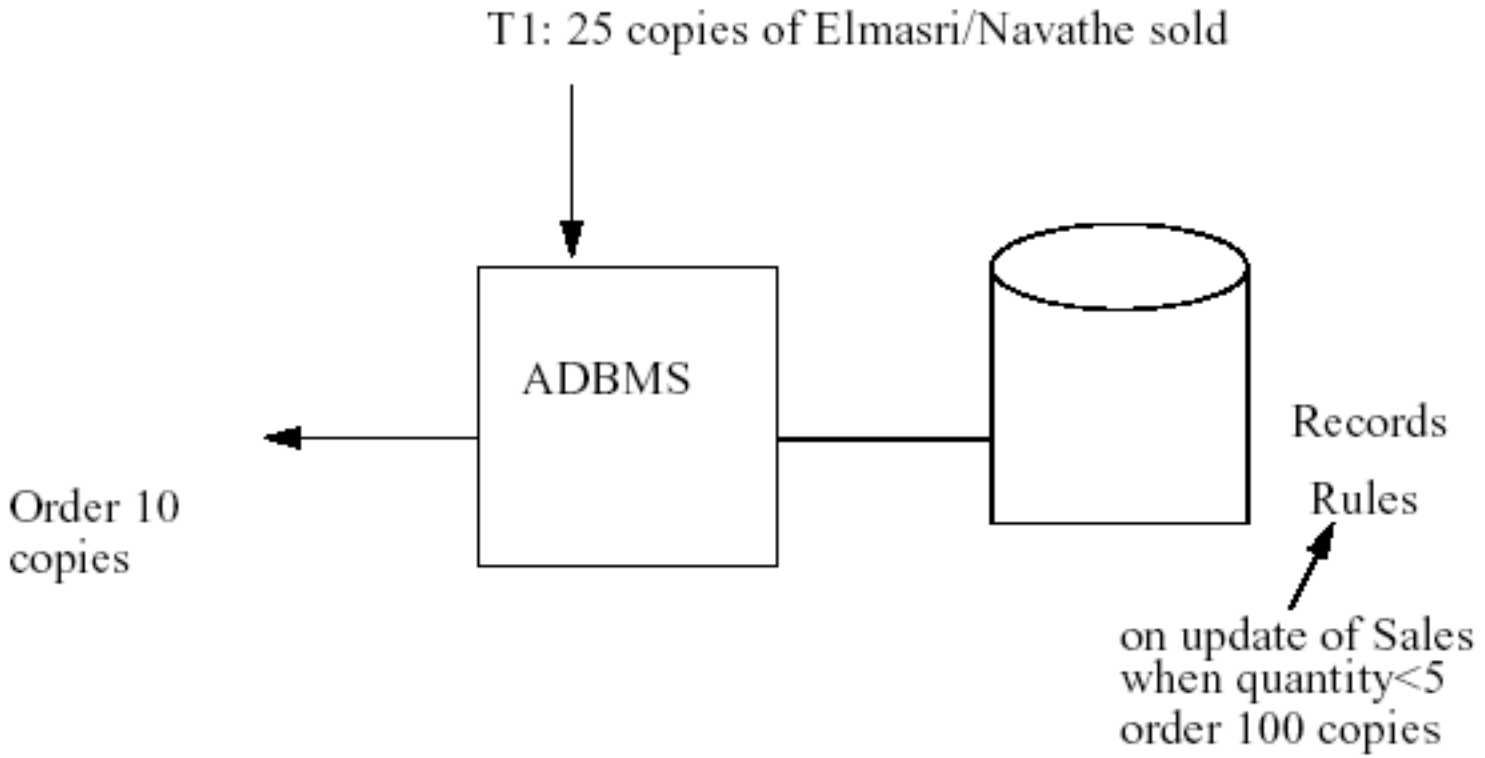


Active Databases

Active DBMS

Recognize predefined situations in database, application, environment

Trigger predefined actions when situations occur, such as DB operations, program executions

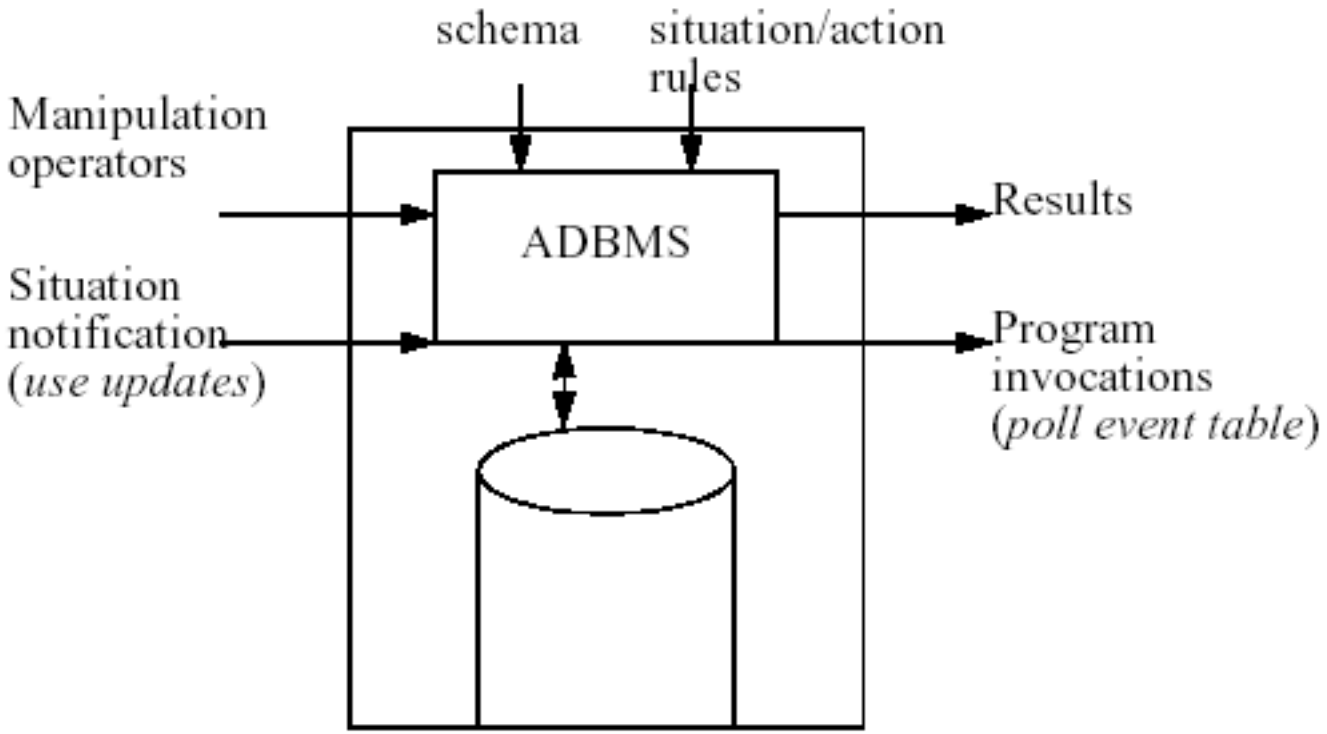


Active Databases

General idea

ADBMS provides:

- Regular DBMS primitives
- + definition of application-defined situations
- + triggering av application-defined reactions



Active Databases

Possible applications of ADBMS:

Consistency enforcement

- Reaction to violations
  - e.g. ROLLBACK when constriant violated
- Connection to time
  - e.g. check some constraint violations every midnight

Computation of derived data

- View materialization of derived data
  - e.g. incremental recomputation of view of sum of salaries per department, salsum(dno,salary), computed from employee(ssn,dno,salary)
- … or invalidation of materialized view when relevant update (e.g. ssalary) occurs => rematerialize view when accessed next time

Automatic travel booking

- Order currently not available ticket with desired properties whever one gets available

Active Databases

Semantics of ECA rules

- Most common model presently
- Event Condition Action:
  - WHEN event occurs
  - IF condition holds
  - DO execute action

Active Databases

Example - Oracle syntax

- EMPLOYEE(SSN, DNO, SALARY)
  DEPT(DNO, MGRSSN)
  SALSUM(DNO, TOTAL) <= Materialized

- ```
  CREATE TRIGGER EMPLOYEE_SALARY_MATERIALIZATION
      AFTER UPDATE ON EMPLOYEE   <--- Event
                                 <--- No condition
      FOR EACH ROW               <--- Action
      BEGIN
         UPDATE SALSUM S
            SET TOTAL = TOTAL - OLD.SALARY
            WHERE S.DNO = OLD.DNO
         UPDATE SALSUM S
            SET TOTAL = TOTAL + NEW.SALARY
            WHERE S.DNO = NEW.DNO
      END;
  ```
  ```

Active Databases (ECA)

Event:
- Update of database record(s)
- Parameterised using pseudo tables OLD and NEW

Condition:
- Query on database state,
- e.g. a database query
  - empty result => condition is FALSE
  - non-empty result => condition is TRUE

Action:
- Database update statement(s)
- Stored procedure execution

Unconditioned (EA) rules, as in example:
- ON ... DO
- Natural in C++/Java based object stires
- Can make arbitrary C++/Java test in DO part!

Condition/Action rules:
- Not common in databases

Active Databases

Example of triggers for constraints, Oracle

```
CREATE TRIGGER SALARY_CONSTRAINT
      AFTER UPDATE OF SALARY ON EMPLOYEE  <--- Event
      WHEN NEW.SALARY >                   <--- Condition
           (SELECT M.SALARY
            FROM EMPLOYEE M,DEPARTMENT D
            WHERE NEW.DNO = D.DNO AND D.MGR = M.SSN)
      BEGIN                               <--- Action
      UPDATE EMPLOYEE E
         SET SALARY = OLD.SALARY
      END;
```

Active Databases

NOTICE! SALARY_CONSTRAINT needed for managers:

```
CREATE TRIGGER SALARY_CONSTRAINT2
      AFTER UPDATE ON EMPLOYEE
      WHEN NEW.SALARY <
           (SELECT E.SALARY
          FROM EMPLOYEE E, DEPARTMENT D
            WHERE E.DNO = D.DNO AND D.MGR = M.SSN)
      BEGIN
         ROLLBACK
      END;
```

NOTICE! SALARY_CONSTRAINT needed for departments too!

NOTICE! SALARY_CONSTRAINT needed for insertions too!

Solution: Integrity constraints!

Active Databases

Advanced level SQL-92 (SYBASE, ORACLE) has assertions too:

```
CREATE ASSERTION SALARY_CONSTRAINT
      CHECK(NOT EXISTS
         (SELECT *
          FROM EMPLOYEE E, EMPLOYEE M,DEPARTMENT D
          WHERE E.SALARY > M.SALARY AND
              E.DNO = D.DNO AND
              D.MGRD = M.SSN))
```

Active Databases

Cautions:

Very powerful mechanism:

- Small statement => massive behaviour changes.
- Rope for programmer.
- Requires careful design
- *Activity design* + traditional database design.

Trace consequences of rule specification/changes:

- Make sure indefinite triggering cannot happen.

Help user design meaningful rules:

- Tool, e.g. simulator

Higher abstraction level needed for ordinary users:

- Assertions.
- Specification of materialized views.
- Graphical tools.

Active Databases

Workflow Control Application

Goal: Control long running activity

- multiple application steps
- may be of long duration
- may be executed by different servers in a distributed system

Issues:

- how to sequence the steps (workflow)?
- if a step fails, committed earlier steps cannot be rolled back: how to compensate?

Approach: use triggers/rules

- model application steps by transactions
- use rules to check constraints, chain steps, and handle exceptions
- flexible response to exceptions, rather tha a fixed compensation policy

Active Databases

SUMMARY

Active DBMSs embed situation-action rules in database

Support many functionalities:

- E.g. Integrity control, access control, monitoring, derived data, change notification

Some ADBMS functionality commercially available as *triggers*: Sybase, Oracle, Interbase, etc.